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Alix Lamouroux

Data Scientist specialized in machine learning, health data, signal processing, and biomedical AI.

Machine Learning ◇ Health Data ◇ Signal Processing ◇ Deep Learning ◇ Medical Devices ◇ Reproducible Research

PROFILE

Data scientist and biomedical signal processing researcher with a PhD focused on clinical neuroimaging, machine learning, and reproducible health-data pipelines. Experienced in developing and validating algorithms on complex physiological signals and medical imaging data, from exploratory analysis to robust implementation. Strong interest in connected health, product-oriented AI, and transforming noisy biomedical data into reliable, actionable health indicators. Fluent in French and English, with experience collaborating across research, engineering, and clinical teams.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher – NeuroAI and Brain Decoding **2026-Now**
CiMeC Center for Mind/Brain Sciences, University of Trento, Italy

Developing machine learning and deep learning models to decode and reconstruct visual content from brain activity.
Working with large-scale neurophysiological datasets and multimodal representations to extract meaningful information from complex biological signals.
Designing reproducible training and evaluation pipelines for brain decoding models, with a focus on robustness, interpretability, and scientific validation.

PhD in Biomedical Signal Processing **2022-2025**
IMT Atlantique, Brest, France

Developed reproducible preprocessing and analysis pipelines for clinical fMRI data in post-stroke rehabilitation and neurofeedback.
Analyzed multimodal biomedical data including EEG and fMRI, with a focus on signal quality, physiological interpretation, functional connectivity, and clinical relevance.
Applied machine learning and statistical modeling to identify biomarkers of recovery and predict behavioral outcomes from health data.
Contributed to AI-oriented projects including EEG-to-speech decoding, EEG foundation models, and multimodal representation learning.
Collaborated with engineers, neuroscientists, and clinicians to translate research questions into validated data-processing and modeling workflows.
Supervisors: Nicolas Farrugia, Pierre Maurel, Giulia Lioi, Julie Coloigner.

R&D Engineer – Medical Imaging and AI **2021-2022**
DESKi, Bordeaux, France

Processed and analyzed medical ultrasound data for musculoskeletal and cardiac applications.
Worked on multimodal registration methods to support image-guided spinal surgery.
Contributed to applied R&D workflows in a startup environment, combining medical constraints, algorithmic development, and product-oriented objectives.

Research Intern – Deep Learning for Echocardiography **2021**
DESKi, Bordeaux, France

Developed deep learning approaches for cardiac ultrasound analysis, including automated assessment of cardiac function.
Worked on medical-image preprocessing, model training, validation, and interpretation in collaboration with R&D engineers and clinical experts.
Supervisors: Olivier Moal and Emilie Roger.

Research Intern – EEG Signal Processing **2020**
IMT Atlantique, Brest, France
Analyzed EEG recordings for the Brain Song project, with a focus on oscillatory activity and experimental signal processing.
Built data-analysis workflows for neural time-series data and contributed to scientific publication.
Supervisors: Nicolas Farrugia and Giulia Lioi.

EDUCATION

Engineering Degree **2019-2021**
IMT Atlantique, Brest, France
Specialization in biomedical signal processing, artificial intelligence, and medical data analysis.

MSc in Applied Mathematics **2018-2019**
Sorbonne Université, Paris, France
Advanced mathematical training with applications to modeling, data analysis, and scientific computing.

Double BSc in Biology and Mathematics **2015-2018**
Sorbonne Université, Roscoff, France | University of Western Australia, Australia
Interdisciplinary training in mathematics, biology, and neuroscience. First two years completed in France and final year completed in Australia.

SKILLS

Machine Learning	Deep learning, supervised learning, representation learning, contrastive learning, model evaluation, feature extraction
Signal Processing	EEG, fMRI, ultrasound, physiological time series, preprocessing pipelines, multimodal data analysis
Programming	Python, PyTorch, TensorFlow, Hugging Face, scikit-learn, NumPy, pandas, MNE, Nipype
Scientific Computing	Reproducible pipelines, statistical analysis, experimental design, validation on clinical and biomedical datasets
Engineering Tools	Git, Docker, Singularity, Linux, collaborative development workflows
Languages	French native, English fluent, Spanish basic, Italian basic

SELECTED MACHINE LEARNING PROJECTS

◇ Investigated the transferability of EEG foundation models for sleep staging and abnormality classification in neonatal EEG recordings.

◇ Applied contrastive learning to EEG and fMRI data to learn shared multimodal representations for brain and health-data analysis.

◇ Trained EEG-to-speech decoding models by mapping neural signals to speech representations using multimodal contrastive approaches.

◇ Developed and released fMRIStroke, an open-source preprocessing pipeline for clinical fMRI data from stroke patients.

◇ Contributed to deep learning methods for automated ejection fraction assessment from 2D cardiac ultrasound.

LEADERSHIP AND INVOLVEMENT

◇ Teaching experience: Foundations of AI, Efficient Deep Learning, Biomedical Signal Processing.

◇ Supervision of research interns on biomedical data analysis and machine learning projects.

◇ Participation in collaborative scientific hackathons and workshops: BrainHack Malta 2022 and EEG workshop in Roscoff.

◇ Vice-President and co-founder of a student sailing association in 2016.

PUBLICATIONS

◇ **How Transferable Are EEG Foundation Models? A Case Study on Sleep Staging** **A. Lamouroux**, L. Benoit, J. Lys, N. Farrugia, G. Lioi. *EUSIPCO*, 2026.

◇ **Neurofeedback-Induced Network Plasticity in Motor and Default Mode Networks Correlates with Motor Recovery After Stroke** **A. Lamouroux**, J. Cologner, P. Maurel, N. Farrugia, S. Butet, I. Bonan, G. Lioi. *Journal of NeuroEngineering and Rehabilitation*, 2026.

◇ **Data-driven Identification of Functional Network Changes in Neurofeedback Stroke Rehabilitation: A Clinical Validation of Network-Based Statistics** **A. Lamouroux**, J. Cologner, P. Maurel, N. Farrugia, I. Bonan, G. Lioi. *EUSIPCO*, 2025.

◇ **Evaluating Lesion-Specific Preprocessing Pipelines for rs-fMRI in Stroke Patients: Impact on Functional Connectivity and Behavioral Prediction** **A. Lamouroux**, J. Cologner, P. Maurel, N. Farrugia, G. Lioi. *Imaging Neuroscience*, 2025.

◇ **fMRIStroke: A Preprocessing Pipeline for fMRI Data from Stroke Patients** **A. Lamouroux**, G. Lioi, J. Cologner, P. Maurel, N. Farrugia. *Journal of Open Source Software*, 2024.

◇ **Explicit and Automatic Ejection Fraction Assessment on 2D Cardiac Ultrasound with a Deep Learning-Based Approach** O. Moal, E. Roger, **A. Lamouroux**, C. Younes, G. Bonnet, B. Moal, S. Lafitte. *Computers in Biology and Medicine*, 2022.

◇ **Beta and Theta Oscillations Correlate with Subjective Time During Musical Improvisation in Ecological and Controlled Settings: A Single Subject Study** N. Farrugia, **A. Lamouroux**, C. Rocher, J. Bouvet, G. Lioi. *Frontiers in Neuroscience*, 2021.

INTERESTS

Sailing ◇ Windsurfing ◇ Trail running ◇ Cycling ◇ Travelling ◇ Hiking